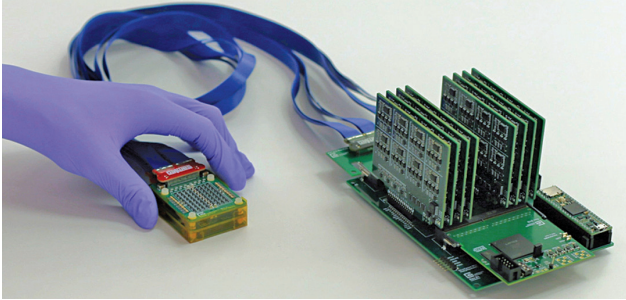


Portable ultrasound sensor for early breast cancer detection

MIT engineers have unveiled a portable ultrasound sensor that could enable earlier breast cancer detection, shrinking bulky imaging systems into a handheld probe and smart-

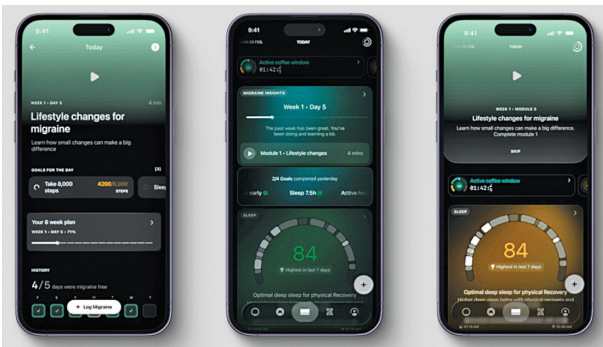


The new system consists of a small ultrasound probe, on left, attached to an acquisition and processing module that is a little larger than a smartphone (Credit: Conformable Decoders Lab at the MIT Media Lab)

phone-sized processor. The miniaturised device produces wide-angle, real-time 3D scans that may help identify tumours between routine mammograms, potentially improving early-stage diagnosis and treatment outcomes. Developed at MIT's Media Lab in the US, the system attaches to a laptop for visualisation and costs around \$300 to build. Researchers are now advancing clinical trials and exploring even smaller, smartphone-linked versions for broader screening use.

Software tool that predict migraines

Migraine PowerPlug is a first-of-its-kind software tool that predicts migraine risk windows before pain starts by analysing biomarkers such as sleep, heart rate variability,



Ultrahuman launches predictive 'Migraine PowerPlug' on a smart ring (Credit: <https://www.clicktherapeutics.com>)

stress and movement. The predictive tool, developed by Ultrahuman and Click Therapeutics as a software add-on for the Ultrahuman Ring AIR, a smart ring, integrates FDA-authorised digital migraine intervention tech to offer personalised lifestyle guidance and early risk alerts. Engineered in collaboration with Click Therapeutics' clinical expertise, the innovation aims to shift migraine care from reactive tracking to proactive management for users worldwide.

Liquid-metal motor ignites electronics innovation

The world's first liquid-metal droplet rotary motor that spins using swirling flows in a metal droplet rather than conventional rigid parts such as coils or magnets has been



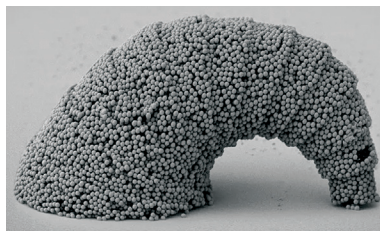
The copper paddle rotates using swirling flows inside a droplet of liquid metal that is embedded in a salt solution and exposed to an electric field (Credit: UNSW)

developed by researchers at the University of New South Wales. The prototype, engineered by a team in Sydney, embeds a liquid-metal droplet in a salt solution and exposes it to an electric field, driving a tiny copper paddle to rotate at up to 320rpm in early

tests. The design's simplicity and inherent flexibility could accelerate advances in soft robotics, flexible electronics, and biomedical devices by enabling motion in systems where traditional motors are impractical.

New way to build small 3D structures

A new fabrication method enables microscopic 3D structures from a broad range of materials, overcoming the polymer-



Croissant-shaped 3D microstructure assembled from silicon dioxide (Credit: <https://interestingengineering.com>)

only limitation of existing techniques. The optofluidic assembly process uses light-induced fluid flows to precisely position particles into complex shapes, which are then freed from a temporary polymer mould. Engineered

by a team at the Max Planck Institute for Intelligent Systems and the National University of Singapore, the technique works with metals, semiconductors, and multifunctional materials, and has produced microvalves and responsive microrobots. This advance broadens design options for microdevices in medicine, robotics, and engineering.

Better conductor for heat management in electronics hardware

A UCLA-led research team has discovered a metallic material with the highest thermal conductivity ever measured, potentially transforming heat management in electronics and AI hardware. Theta-phase tantalum nitride, engineered by scientists at the UCLA Samueli School of Engineering, conducts heat at about 1100W/m·K, nearly three times that of copper and silver, which have long been the industry benchmarks.